Examiner reports

Q1.

This question was generally done very well apart from part (c)(ii).

In part (a), most candidates did the two expansions accurately and clearly, with the common error being not to raise the 3 to the same power as x.

For part (b), most candidates knew what was required and used an appropriate technique, either substituting values of x or setting up simultaneous equations. A few candidates added in a constant, which was acceptable if they showed it was zero. Errors tended to be algebraic and/or numerical.

For part (c)(i), most candidates knew that they were supposed to use their result from part (b), although some did not indicate this and started with an incorrect expression, having inverted their coefficients, and thus scored no marks. Another error often seen here was to raise $(1 + x)$ and $(1 + 3x)$ to the power of the coefficients found in part (b).A few candidates tried to use their expansions in place of the denominator. An algebraic slip in simplification cost many otherwise successful candidates the final mark. A few candidates did not use the partial fractions and wrote the expression as a product using their expansions; this is quite acceptable but involves a lot more work to achieve the correct answer.

In part (c)(ii), most candidates seemed unfamiliar with defining a valid range for a binomial expansion. Some attempted to use the modulus notation, but errors such as $|x| < -3$ showed a lack of understanding of it. Some more successful candidates only gave half the range, omitting the negative numbers, and the inequality often went the wrong way.

Many candidates argued that $x \neq -1$ and $x \neq -\frac{1}{3}$ defined the range as you cannot divide by zero.

Q2.

This question was usually done very well, with many candidates scoring highly. Binomial expansions and use of partial fractions were seen to be generally well known with most of the algebraic manipulation being done accurately. Most knew the result for part (a) (i) although some made errors with the sign of the x term and lost both marks. In part (ii) most made the expected start of $\frac{1}{3}(1 - \frac{2}{3}x)^{-1}$ and then gave a detailed expansion rather than use the result from part (i) with $\frac{2}{3}x$ replacing x in their expansion. Some candidates had 3 where $\frac{1}{3}$ should be, it just 'becoming' $\frac{1}{3}$ to get the given answer. A few candidates manipulated powers of 2 and 3 for no good reason other than to achieve the given answer, most unconvincingly. Some used the expansion of $(a + bx)^n$ from the formula book, often successfully but the few who attempted to use $\binom{n}{r}$ with $n = -1$ gained no credit. The response to part (b) was similar to that in part (a) (i).

Most candidates knew how to start part (c) although the accuracy of the algebraic expression written down often left something to be desired. Those who substituted x = 1, $x = \frac{3}{2}$ and x = 0 were usually more successful in getting A,B and C correctly than those who multiplied out and set up simultaneous equations by equating coefficients. Solutions to these equations were often just written down with no working seen. This gained full credit if they were correct, but usually none otherwise. Most attempted part (d) using the results from previous parts of the question. Many did this correctly although some made algebraic or numerical slips and some candidates used the reciprocals of A, B and C. A few did not use their partial fraction expression but multiplied out the product of the series directly often resulting in an algebraic error.

Q3.

The binomial expansion continues to cause candidates problems. Many candidates made a correct start to part (a) but then poor manipulation with signs and powers of 2 led to incorrect work. Some candidates quoted the formulae printed on page 4 of the formulae booklet but then showed a lack of understanding as '1.2' was interpreted as '1 $\frac{2}{10}$' rather than '1 × 2'. Only the more able candidates were able to deal correctly with parts (b) and (c) with many others making the errors '$(2 + x)^9 = 2(1 + \frac{x}{2})^9$', '$(2 + x)^9 = 2(1 + x)^9$', or even, in part (c), ignoring the 2 completely. Some able candidates wasted time by finding the complete expansions in parts (b) and (c).

Q4.

Part (a) There were many errors made in both signs and coefficients in both the x and x^2 term. The common errors were $-3x$ and $3x$, the latter from $\frac{(-3 \times -2)}{2}$. A coefficient of 10 was also seen, from 12/1.2, where the factorial had been misunderstood.

Part (b) Here the expectation was that candidates would replace x in their answer to (i) with $\frac{5}{2}x$ and many did so, whereas others just started again ignoring the 'hence'. Their answer was often inconsistent with part (a). A common error was poor notation, such as missing brackets in $\left(\pm \frac{5}{2}x\right)^2$ or failure to square the $\frac{5}{2}$. Those who did the latter, and showed no working, received no credit.

Part (c) Relatively few candidates could quote the range correctly, although most knew it was something 5 2 to do with ±5/2 or ±2/5 which were used in an inequality or with a not equal to sign. Those who tried to 2 5 work it out starting from $|x| < 1$ were generally more successful.

Part (d) Although some excellent answers were seen, relatively few candidates could carry out the required manipulation of the indices correctly. Most realised they were to use their answer to part (b) but gave answers in which a, b and c were not integers, without apparently noting the fact. However, there were incidences of poor algebra, such $\left(\frac{4}{2-5x}\right) = 2 - \frac{4}{5x}$ whilst others managed to manipulate the given expression into a quadratic expansion.

Q5.

Use of the binomial expansion and use of the expansion of $(2 + x)(2 + x)^2$ were equally popular in part (a)(i) and most obtained the correct answer. There was however a minority of candidates who did not include the powers of 2 in the middle terms when applying the binomial expansion. It was disappointing to see so many candidates producing masses of algebra rather than replacing x by $(-x)$ to obtain the answer to part (a)(ii). There was some evidence of 'faking' to reach the printed answer in part (b), which quite often resulted in marks being lost as earlier correct work was changed. The vast majority realised that $\mathrm{d}y/\mathrm{d}x$ was required in part (c), although some later explanations could have been more precise.

Q6.

Many candidates were able to give the correct expansion in part (a), normally by applying Pascal's triangle. Many realised that x should be replaced by reaching the exact printed answer. Candidates should be aware that approximate decimal values for each of 4 $\sqrt{5}$ and $4\left(\sqrt{5}\right)^3$ cannot then be added and the result used to equate to the exact surd part of the printed answer. The final part of the question proved difficult, although, again, some elegant solutions were presented. Those who wrote '$\log_2(1 + \sqrt{5})^4 = \log_2(56 + 24\sqrt{5}) = \log_2[8(7 + 3\sqrt{5})]$' usually went on to score all three marks.

Q7.

Part (a) Most candidates were successful here, using the formula book to help them with the expansion if necessary. There was some inaccuracy in the x^2 term.

Part (b) Most candidates knew they were requested to expand $(1 + 2x)^{-\frac{1}{2}}$ and attempted to use their result from part (a). The common error was to use $2x^2$ for $(2x)^2$ although some left the coefficient of the x^2 term as 12/8 without further simplification.

Part (c) Most candidates substituted $x = -0.1$into their expansion from part (b) and then many who had obtained 1.115 just doubled it to obtain the given answer, apparently believing they had answered the 1_5question. Few candidates convincingly linked $(0.8)^{-\frac{1}{2}}$ with 5/4 and so answered the question.